MODULE 3
PRINCIPLES OF MEASUREMENT

3.1 Static Characteristics

The static characteristics of an instrument are, in general, considered for instruments which are used to measure an unvarying process condition. All the static performance characteristics are obtained by one form or another of a process called calibration. There are a number of related definitions (or characteristics), which are described below, such as accuracy, precision, repeatability, resolution, errors, sensitivity, etc.

1. **Instrument:** A device or mechanism used to determine the present value of the quantity under measurement.
2. **Measurement:** The process of determining the amount, degree, or capacity by comparison (direct or indirect) with the accepted standards of the system units being used.
3. **Accuracy:** The degree of exactness (closeness) of a measurement compared to the expected (desired) value.
4. **Resolution:** The smallest change in a measured variable to which an instrument will respond.
5. **Precision**: A measure of the consistency or repeatability of measurements, i.e. successive reading do not differ. (Precision is the consistency of the instrument output for a given value of input).
6. **Expected value:** The design value, i.e. the most probable value that calculations indicate one should expect to measure.
7. **Error:** The deviation of the true value from the desired value.
8. **Sensitivity:** The ratio of the change in output(response) of the instrument to a change of input or measured variable.

3.2 Error in Measurement

Measurement is the process of comparing an unknown quantity with an accepted standard quantity. It involves connecting a measuring instrument into the system under consideration and observing the resulting response on the instrument. The measurement thus obtained is a quantitative measure of the so-called "true value" (since it is very difficult to define the true value, the term "expected value" is used). Any measurement is affected by many variables; therefore the results rarely reflect the expected value. For example, connecting a measuring instrument into the circuit under consideration always disturbs (changes) the circuit, causing the measurement to differ from the expected value. Some factors that affect the measurements are related to the measuring instruments themselves. Other factors are related to the person using the instrument. The degree to which a measurement nears the expected value is expressed in terms of the error of measurement. Error may be expressed either as absolute or as percentage of error. Absolute error may be defined as the difference between the expected value of the variable and the measured value of the variable.

$$e = Y_n - X_n$$

where e = absolute error
Yn = expected value
Xn = measured value
Therefore,

$$\%\ \text{Error} = \frac{\text{Absolute value}}{\text{Expected value}} \times 100 = \frac{e}{Y_n} \times 100$$

$$\%\ \text{Error} = \left(\frac{Y_n - X_n}{Y_n}\right) \times 100$$

It is more frequently expressed as a accuracy rather than error.
Therefore,

$$A = 1 - \left| \frac{Y_n - X_n}{Y_n} \right|$$

where A is the relative accuracy.
Accuracy is expressed as % accuracy.

$$a = 100\% - \%\text{ error}$$

$$a = A \times 100\ \%$$

where a is the % accuracy.

Example 1

The expected value of the voltage across a resistor is 80 V. However, the measurement gives a value of 79 V. Calculate (i)absolute error (ii)% error (iii)relative accuracy (iv)% of accuracy

Solution:

$$\text{Absolute error } e = Y_n - X_n = 80 - 79 = 1\text{ V}$$

$$\%\text{ Error} = \frac{Y_n - X_n}{Y_n} \times 100 = \frac{80 - 79}{80} \times 100 = 1.25\%$$

Relative Accuracy,

$$A = 1 - \left| \frac{Y_n - X_n}{Y_n} \right| = 1 - \left| \frac{80 - 79}{80} \right|$$

$$A = 1 - 1/80 = 79/80 = 0.9875$$

% of Accuracy,

$$a = 100 \times A = 100 \times 0.9875 = 98.75\%$$

$$a = 100\% - \%\text{ of error} = 100\% - 1.25\% = 98.75\%$$

Example 2

The expected value of the current through a resistor is 20 mA. However the measurement yields a current value of 18mA. Calculate (i)absolute error (ii)% error (iii)relative accuracy (iv)% accuracy

Solution:

Absolute error,

$$e = Y_n - X_n$$

where e = error, Yn = expected value, Xn = measured value
Given Yn = 20 mA and Xn = 18 mA
Therefore e = Yn – Xn = 20 mA – 18 mA = 2 mA
% error

$$\%\text{ error} = \frac{Y_n - X_n}{Y_n} \times 100 = \frac{20\text{ mA} - 18\text{ mA}}{20\text{ mA}} \times 100 = \frac{2\text{ mA}}{20\text{ mA}} \times 100 = 10\%$$

Relative accuracy

$$A = 1 - \left| \frac{Y_n - X_n}{Y_n} \right| = 1 - \left| \frac{20\text{ mA} - 18\text{ mA}}{20\text{ mA}} \right| = 1 - \frac{2}{20} = 1 - 0.1 = 0.90$$

% accuracy
a = 100% – %error = 100% – 10% = 90%
a = A x 100% = 0.90 x 100% = 90%

If a measurement is accurate, it must also be precise, i.e. Accuracy means precision. However, a precision measurement may not be accurate. (The precision of a measurement is a quantitative or numerical indication of the closeness with which a repeated set of measurement of the same variable agree with the average set of measurements.) Precision can also be expressed mathematically as

$$P = 1 - \left| \frac{X_n - \bar{X}_n}{\bar{X}_n} \right|$$

where Xn = value of the nth measurement
Xn bar = average set of measurement

Example 3
Table gives the set of 10 measurement that were recorded in the laboratory. Calculate the precision of the 6th measurement.

Measurement Number	1	2	3	4	5	6	7	8	9	10
Measurement Value	98	101	102	97	101	100	103	98	106	99

Solution:
The average value for the set of measurements is given by

$$\bar{X}_n = \frac{\text{Sum of the 10 measurement values}}{10}$$

$$= \frac{1005}{10} = 100.5$$

$$\text{Precision} = 1 - \left| \frac{X_n - \bar{X}_n}{\bar{X}_n} \right|$$

For the 6th reading

$$\text{Precision} = 1 - \left| \frac{100 - 100.5}{100.5} \right| = 1 - \frac{0.5}{100.5} = \frac{100}{100.5} = 0.995$$

The accuracy and precision of measurements depend not only on the quality of the measuring instrument but also on the person using it. However, whatever the quality of the instrument and the case exercised by the user, there is always some error present in the measurement of physical quantities.

3.3 Types of Static Error

The static error of a measuring instrument is the numerical difference between the true value of a quantity and its value as obtained by measurement, i.e. repeated measurement of the same quantity gives different indications. Static errors are categorised as gross errors or human errors, systematic errors, and random errors.

3.3.1 Gross Errors

These errors are mainly due to human mistakes in reading or in using instruments or errors in recording observations. Errors may also occur due to incorrect adjustment of instruments and computational mistakes. These errors cannot be treated mathematically. The complete elimination of gross errors is not possible, but one can minimize them.

Some errors are easily detected while others may be elusive. One of the basic gross errors that occurs frequently is the improper use of an instrument. The error can be minimized by taking proper care in reading and recording the measurement parameter. In general, indicating instruments change ambient conditions to some extent when connected into a complete circuit. One should therefore not be completely dependent on one reading only. At least three separate readings should be taken, preferably under conditions in which instruments are switched off and on.

3.3.2 Systematic Errors

These errors occur due to shortcomings of the instrument, such as defective or worn parts, or ageing or effects of the environment on the instrument. These errors are sometimes referred to as bias, and they influence all measurements of a quantity alike. A constant uniform deviation of the operation of an instrument is known as a systematic error. There are basically three types of systematic errors:
(i) Instrumental, (ii) Environmental, and (iii) Observational.

(i) Instrumental Errors

Instrumental errors are inherent in measuring instruments, because of their mechanical structure. For example, in the D' Arsonval movement, friction in the bearings of various moving components, irregular spring tensions, stretching of the spring, or reduction in tension due to improper handling or overloading of the instrument.
Instrumental errors can be avoided by
(a) selecting a suitable instrument for the particular measurement applications.
(b) applying correction factors after determining the amount of instrumental error.
(c) calibrating the instrument against a standard.

(ii) Environmental Errors

Environmental errors are due to conditions external to the measuring device, including conditions in the area surrounding the instrument, such as the effects of change in temperature, humidity, barometric pressure or of magnetic or electrostatic fields. These errors can also be avoided by (i) air conditioning, (ii) hermetically sealing certain components in the instruments, and (iii) using magnetic shields.

(iii) Observational Errors

Observational errors are errors introduced by the observer. The most common error is the parallax error introduced in reading a meter scale, and the error of estimation when obtaining a reading from a meter scale. These errors are caused by the habits of individual observers. For example, an observer may always introduce an error by consistently holding his head too far to the left while reading a needle and scale reading. In general, systematic errors can also be subdivided into static and dynamic errors. Static errors are caused by limitations of the measuring device or the physical laws governing its behaviour. Dynamic errors are caused by the instrument not responding fast enough to follow the changes in a measured variable.

Example 4

A voltmeter having a sensitivity of 1 kΩ/V is connected across an unknown resistance in series with a milliammeter reading 80 V on 150 V scale. When the milliammeter reads 10 mA, calculate the (i) Apparent resistance of the unknown resistance, (ii) Actual resistance of the unknown resistance, and (iii) Error due to the loading effect of the voltmeter.

Solution:
The total circuit resistance,

$$R_T = \frac{V_T}{I_T} = \frac{80}{10\ \text{mA}} = 8\ \text{k}\Omega$$

(Neglecting the resistance of the milliammeter.)
The voltmeter resistance equals,

$$R_v = 1000\ \Omega/\text{V} \times 150 = 150\ \text{k}\Omega$$

Actual value of unknown resistance,

$$R_x = \frac{R_T \times R_v}{R_v - R_T} = \frac{8\,\text{k} \times 150\,\text{k}}{150\,\text{k} - 8\,\text{k}}$$

$$= \frac{1200\,\text{k}^2}{142\,\text{k}} = 8.45\ \Omega$$

$$\%\ \text{error} = \frac{\text{Actual value} - \text{Apparent value}}{\text{Actual value}} = \frac{8.45\,\text{k} - 8\,\text{k}}{8.45\,\text{k}} \times 100$$

= 0.053 x 100 = 5.3%

Example 5
Referring to Example 4, if the milliammeter reads 600 mA and the voltmeter reads 30 V on a 150 V scale, calculate the following:
(i)Apparent, resistance of the unknown resistance. (ii) Actual resistance of the unknown resistance. (iii) Error due to loading effect of the voltmeter. Comment on the loading effect due to the voltmeter for both Examples 4 and 5. (Voltmeter sensitivity given 1000 Ω/V)
Solution:
The total circuit resistance is given by,

$$R_T = \frac{V_T}{I_T} = \frac{30}{0.6} = 50\ \Omega$$

The voltmeter resistance Rv equals
Rv = 1000 Ω / V x 150 = 150 kΩ
Neglecting the resistance of the milliammeter, the value of unknown resistance = 50 Ω

$$R_x = \frac{R_T \times R_v}{R_v - R_T} = \frac{50 \times 150\ \text{k}}{150\ \text{k} - 50} = \frac{7500\ \text{k}}{149.95\ \text{k}} = 50.0167\ \Omega$$

$$\%\ \text{Error} = \frac{50.0167 - 50}{50.0167} \times 100 = \frac{0.0167}{50.0167} \times 100 = 0.033\%$$

In Example 4, a well calibrated voltmeter may give a misleading resistance when connected across two points in a high resistance circuit. The same voltmeter, when connected in a low resistance circuit (Example 5) may give a more dependable reading. This show that voltmeters have a loading effect in the circuit during the measurement.

3.3.3 Random Errors

These are errors that remain after gross and systematic errors have been substantially reduced or at least accounted for. Random errors are generally an accumulation of a large number of small effects and may be of real concern only in measurements requiring a high degree of accuracy. Such errors can be analysed statistically. These errors are due to unknown causes, not determinable in the ordinary process of making measurements. Such errors are normally small and follow the laws of probability. Random errors can thus be treated mathematically.

For example, suppose a voltage is being monitored by a voltmeter which is read at 15 minutes intervals. Although the instrument operates under ideal environmental conditions and is accurately calibrated before measurement, it still gives readings that vary slightly over the period of observation. This variation cannot be corrected by any method of calibration or any other known method of control.

3.4 Sources of Error

The sources of error, other than the inability of a piece of hardware to provide a true measurement, are as follows:
1. Insufficient knowledge of process parameters and design conditions
2. Poor design
3. Change in process parameters, irregularities, upsets, etc.
4. Poor maintenance
5. Errors caused by person operating the instrument or equipment
6. Certain design limitations

DIGITAL VOLTMETERS

3.5 Introduction

Digital Voltmeters(DVMs) are measuring instruments that convert analog voltage signals into a digital or numeric readout. This digital readout can be displayed on the front panel and also used as an electrical digital output signal.
Any DVM is capable of measuring analogue dc voltages. However, with appropriate signal conditioners preceding the input of the DVM, quantities such as AC voltages, ohms, DC and AC current, temperature and pressure can be measured. The common element in all these signals conditioners is the DC voltage, which is proportional to the level of the unknown quantity being measured. This DC output is then measured by the DVM.

DVMs have various features such as speed, automation operation and program ability. There are several varieties of DVM which differ in the following ways:
- Number of digits
- Number of measurements
- Accuracy
- Speed of reading
- Digital output of several types

The DVM displays AC and DC voltages as discrete numbers rather than as a pointer on continuous scale as in an analog voltmeter. A numerical readout is advantageous because it reduces human error, eliminates parallax error, increases reading speed and often provides output in digital form suitable for further processing and recording. With the development of IC modules, the size, power requirements and cost of DVMs have been reduced. DVMs compete with analogue voltmeters in portability and size.
Their outstanding qualities are their operating and performance characteristics are given below:
1. Input range from +1.000 V to +1000 V with automatic range selection and overload indication.
2. Absolute accuracy as high as ± 0.005% of the reading.
3. Resolution 1 part in million (1 μV reading can be read or measured on 1 V range).
4. Input resistance typically 10 MΩ, input capacitance 40 pF.
5. Calibration internally from stabilised reference sources, independent of measuring circuit.
6. Output in BCD form, for print output and further digital signal processing. Optional features may include additional circuitry to measure current, ohms and voltage ratio.

3.6 Ramp Technique

The operating principle is to measure the time that a linear ramp takes to change the input level to the ground level, or vice-versa. This time period is measured with an electronic time-interval counter and the count is displayed as a number of digits on an indicating tube or display. The operating principle and block diagram of a ramp type DVM are shown in Figures 3.1 and 3.2.

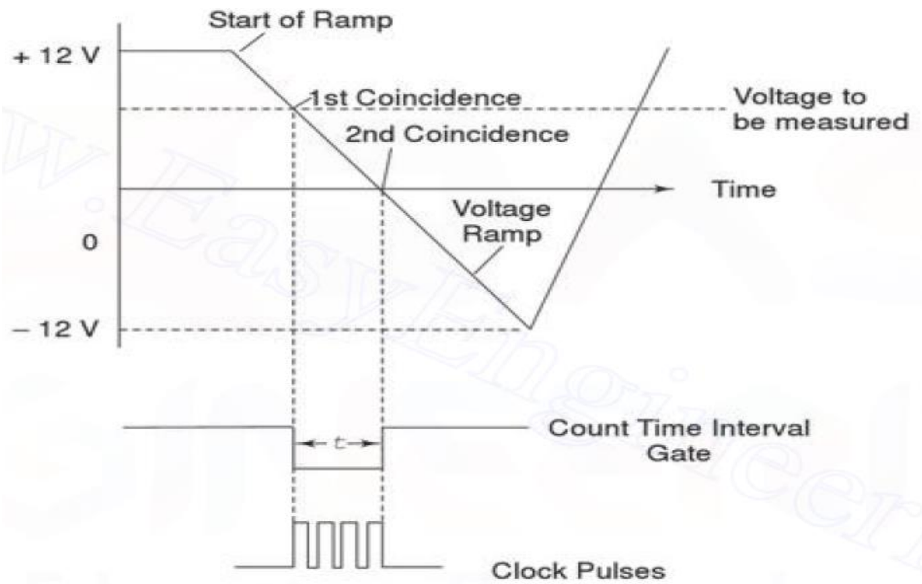

Figure 3.1 Voltage to Time Conversion

The ramp may be positive or negative; in this case a negative ramp has been selected. At the start of the measurement a ramp voltage is initiated (counter is reset to 0 and sampled rate multivibrator gives a pulse which initiates the ramp generator). The ramp voltage is continuously compared with the voltage that is being measured. At the instant these two voltage become equal, a coincidence circuit generates a pulse which opens a gate, i.e. the input comparator generates a start pulse. The ramp continues until the second comparator circuit senses that the ramp has reached zero value. The ground comparator compares the ramp with ground. When the ramp voltage equals zero or reaches ground potential, the ground comparator generates a stop pulse. The output pulse from this comparator closes the gate. The time duration of the gate opening is proportional to the input voltage value. In the time interval between the start and stop pulses, the gate opens and the oscillator circuit drives the counter. The magnitude of the count indicates the magnitude of the input voltage, which is displayed by the readout. Therefore, the voltage is converted into time and the time count represents the magnitude of the voltage. The sample rate multivibrator determines the rate of cycle of measurement. A typical value is 5 measuring cycles per second, with an accuracy of ± 0.005% of the reading. The sample rate circuit provides an initiating pulse for the ramp generator to start its next ramp voltage. At the same time a reset pulse is generated, which resets the counter to the zero state. Any DVM has a fundamental cycle sequence which involves sampling, displaying and reset sequences.

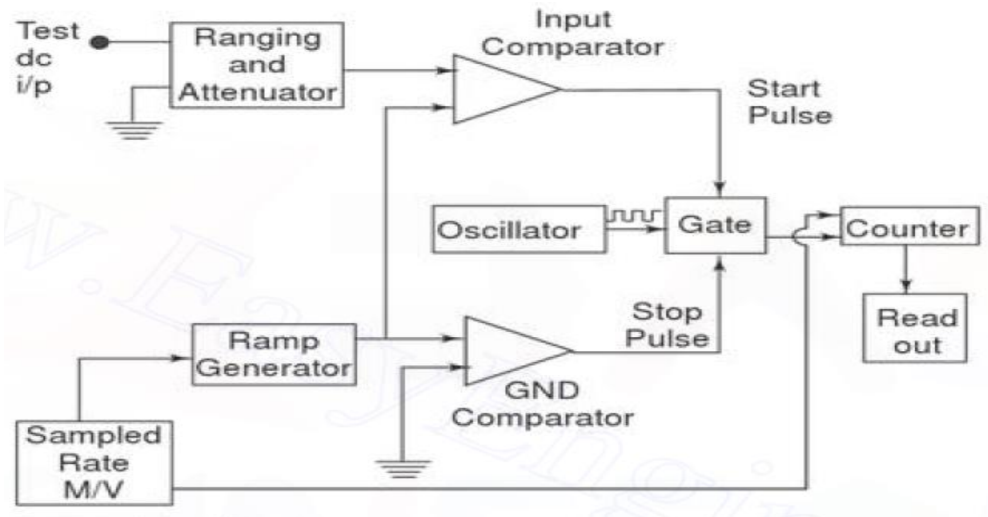

Figure 3.2 Block Diagram of Ramp Type DVM

Advantages

- The Ramp technique circuit is easy to design.
- Its cost is low.
- The output pulse can be transmitted over long feeder lines.

Disadvantages

- Single ramp requires excellent characteristics regarding linearity of the ramp and time measurement.
- Large errors are possible when noise is superimposed on the input signal.
- Input filters are usually required with this type of converter.

3.7 Dual Slope Integrating Type DVM (Voltage to Time conversion)

In ramp techniques, superimposed noise can cause large errors. In the dual ramp technique, noise is averaged out by the positive and negative ramps using the process of integration.

3.7.1 Principle of Dual Slope Type DVM

As illustrated in Figure 3.3, the input voltage 'e_i' is integrated, with the slope of the integrator output proportional to the test input voltage. After a fixed time, equal to t_1, the input voltage is disconnected and the integrator input is connected to a negative voltage $-e_r$. The integrator output will have a negative slope which is constant and proportional to the magnitude of the input voltage. The block diagram is given in Figure 3.4.

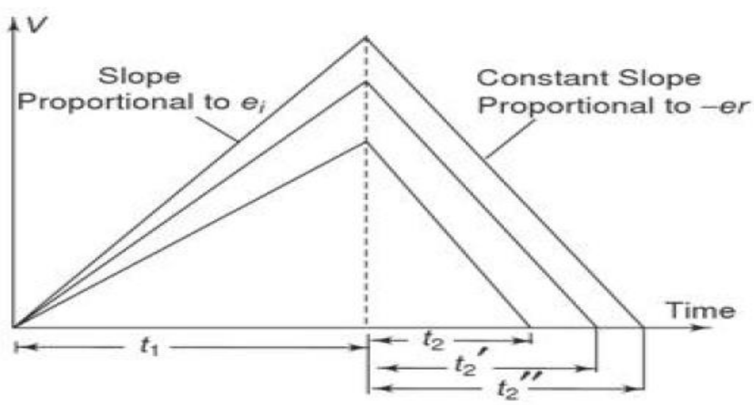

Figure 3.3 Basic Principle of Dual Slope Type DVM

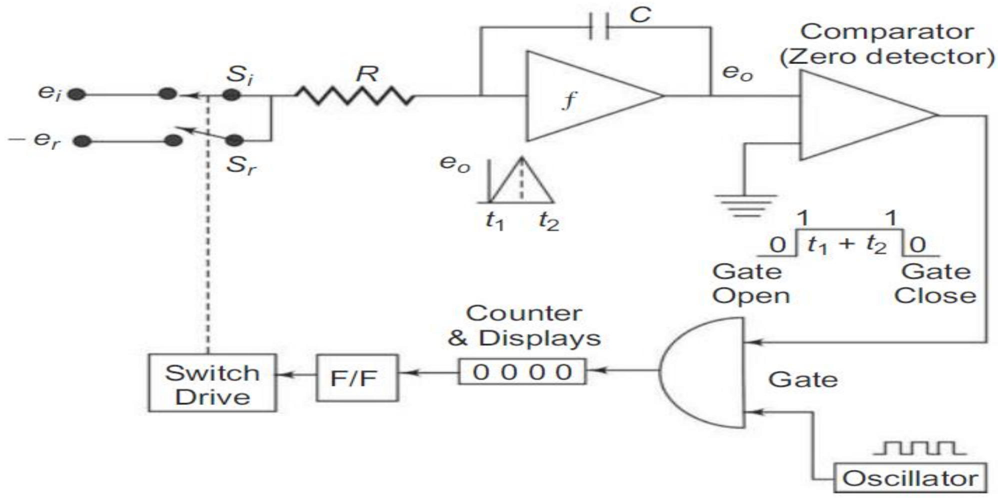

Figure 3.4 Block Diagram of a Dual Slope Type DVM

At the start a pulse resets the counter and the F/F output to logic level '0'. Si is closed and Sr is open. The capacitor begins to charge. As soon as the integrator output exceeds zero, the comparator output voltage changes state, which opens the gate so that the oscillator clock pulses are fed to the counter. (When the ramp voltage starts, the comparator goes to state 1, the gate opens and clock pulse drives the counter.) When the counter reaches maximum count, i.e. the counter is made to run for a time 't_1' in this case 9999, on the next clock pulse all digits go to 0000 and the counter activates the F/F to logic level '1'. This activates the switch drive, ei is disconnected and – er is connected to the integrator. The integrator output will have a negative slope which is constant, i.e. integrator output now decreases linearly to 0 volts. Comparator output state changes again and locks the gate. The discharge time t2 is now proportional to the input voltage. The counter indicates the count during time t2. When the negative slope of the integrator reaches zero, the comparator switches to state 0 and the gate closes, the capacitor C is now discharged with a constant slope. As soon as the comparator input (zero detector) finds that e_o is zero, the counter is stopped. The pulses counted by the counter thus have a direct relation with the input voltage.

During charging,

$$e_o = -\frac{1}{RC}\int_0^t e_i\,dt = -\frac{e_i\ t_1}{RC} \qquad \text{-----------(3.1)}$$

During discharging,

$$e_o = \frac{1}{RC}\int_0^{t_2} -e_r.dt = -\frac{e_r.\ t_2}{RC} \qquad \text{---------(3.2)}$$

Subtracting Equation 3.1 from 3.2 we have,

$$e_o - e_o = \frac{-e_r\ t_2}{RC} - \left(\frac{-e_i\ t_1}{RC}\right)$$

$$0 = \frac{-e_r.\ t_2}{RC} - \left(\frac{-e_i\ t_1}{RC}\right)$$

$$\frac{e_r.\ t_2}{RC} = \frac{e_i\ t_1}{RC}$$

$$e_i = e_r \frac{t_2}{t_1}$$ ---------(3.3)

If the oscillator period equals T and the digital counter indicates n_1 and n_2 counts respectively,

$$e_i = \frac{n_2 T}{n_1 T} e_r$$

i.e. $$e_i = \frac{n_2}{n_1} e_r$$

Now, n_1 and e_r are constants.
Let

$$K_1 = \frac{e_r}{n_1}$$

Then

$$e_i = K_1 \, n_2$$ ---------(3.4)

From Equation 3.3 it is evident that the accuracy of the measured voltage is independent of the integrator time constant. The times t_1 and t_2 are measured by the count of the clock given by the numbers n_1 and n_2 respectively. The clock oscillator period equals T and if n_1 and er are constants, then Equation 3.4 indicates that the accuracy of the method is also independent of the oscillator frequency. The dual slope technique has excellent noise rejection because noise and superimposed ac are averaged out in the process of integration. The speed and accuracy are readily varied according to specific requirements. Also, an accuracy of ± 0.05% in 100ms is available.

3.8 Integrating Type DVM (Voltage to Frequency conversion)
3.8.1 Principle of Operation

The principle of operation of an integrating type DVM is illustrated in Figure 3.5. A constant input voltage is integrated and the slope of the output ramp is proportional to the input voltage. When the output reaches a certain value, it is discharged to 0 and another cycle begins. The frequency of the output waveform is proportional to the input voltage. The block diagram is illustrated in Figure 3.6.

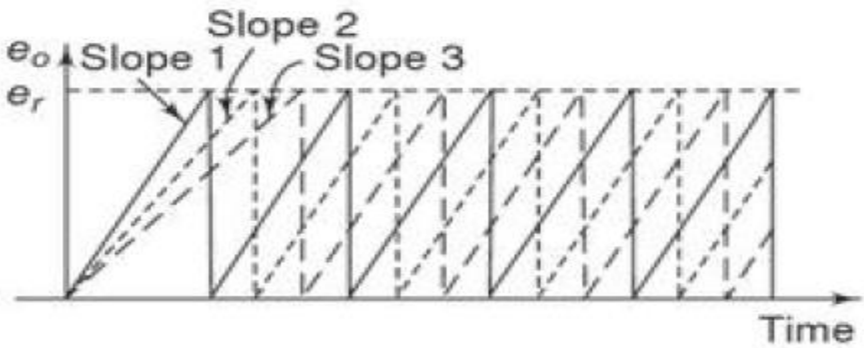

Figure 3.5 Voltage to Frequency conversion

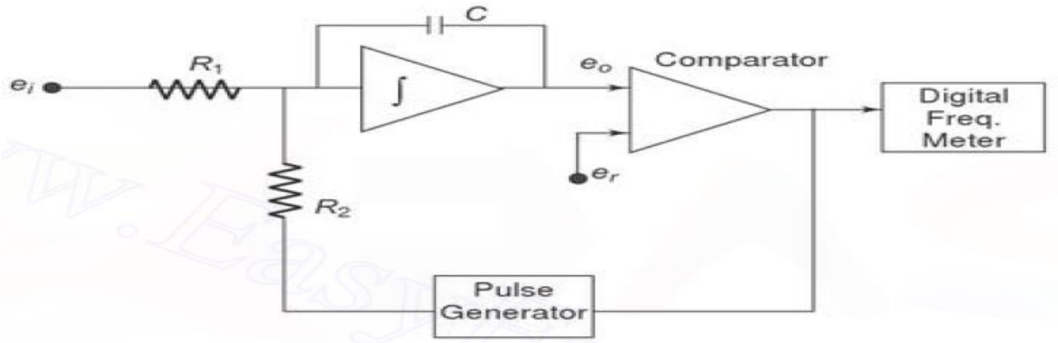

Figure 3.6 Block Diagram of an Integrating Type DVM

The input voltage produces a charging current, e_i/R_1, that charges the capacitor C to the reference voltage e_r. When e_r is reached, the comparator changes state, so as to trigger the precision pulse generator. The pulse generator produces a pulse of precision charge content that rapidly discharges the capacitor. The rate of charging and discharging produces a signal frequency that is directly proportional to e_i.

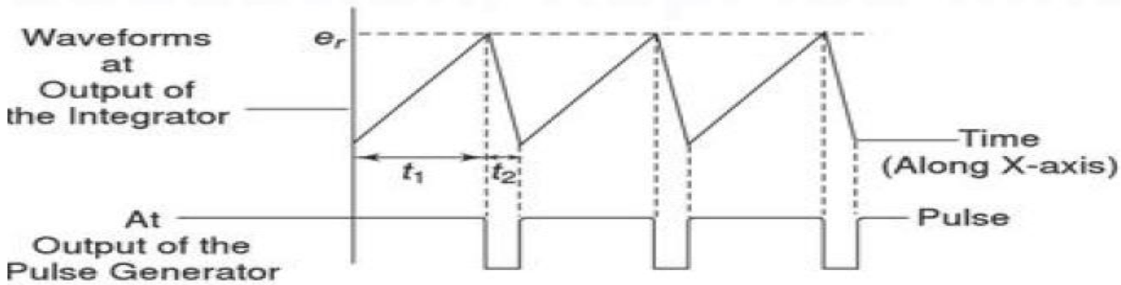

Figure 3.7 Waveforms at the output of the Integrator

The voltage-frequency conversion can be considered to be a dual slope method, as shown in Figure 3.7

Referring to Equation 3.3 we have,

$$e_i = \frac{e_r . t_2}{t_1}$$

But in this case e_r and t_2 are constants.

Let

$$K_2 = e_r . t_2$$

$$e_i = K_2 \left(\frac{1}{t_1} \right) = K_2 (f_0)$$

The output frequency is proportional to the input voltage e_i. This DVM has the disadvantage that it requires excellent characteristics in linearity of the ramp. The ac noise and supply noise are averaged out.

Example 1

An integrator contains 100 kΩ and 1 μF capacitor. If the voltage applied to the integrator output is 1 V, what voltage will be present at the output of the integrator after 1s?

Solution:

using the equation,

$$e_o = \frac{e_i \times t_1}{RC} = \frac{1 \times 1\text{ s}}{100\text{ k} \times 1\,\mu\text{F}} = \frac{1}{0.1} = 10\text{ V}$$

Example 2
Now if a reference voltage is applied to the integrator of the above example, at time t_1 is 5 V in amplitude, what is the time interval of t_2?
Solution:
using the equation,
$$\frac{e_i \times t_1}{RC} = \frac{e_r \times t_2}{RC}$$
Therefore,
$$t_2 = \frac{e_i}{e_r} \times t_1$$
$$t_2 = \frac{1 \times 1}{5} = 0.2 \text{ s}$$

Example 3
An integrator consists of a 100 kΩ and 2 μF capacitor. If the applied voltage is 2 V, what will be the output of the integrator after 2 seconds?
Solution:
Given, R = 100 kΩ, C = 2 μF, e_1 = 2 V and t_1 = 2 s
Using the equation,
$$e_0 = \frac{e_1 \times t_1}{R \times C} = \frac{2\text{ V} \times 2\text{ s}}{100\text{ k} \times 2\,\mu\text{F}} = \frac{4}{200 \times 10^3 \times 10^{-6}} = \frac{4 \times 10^3}{200} = \frac{4000}{200} = 20\text{ V}$$

Example 4
Now if a reference voltage of 10 V is applied to the integrator of the above example at time t_1, what is the time interval of t_2?
Solution:
Given reference voltage 10 V,
$$\frac{e_1 \times t_1}{R \times C} = \frac{e_2 \times t_2}{R \times C}$$
Therefore,
$$t_2 = \frac{e_1 \times t_1}{e_2} = \frac{2 \times 2\text{ s}}{10} = 0.4\text{ s}$$

3.9 Successive Approximations
The successive approximation principle can be easily understood using a simple example, the determination of the weight of an object. By using a balance and placing the object on one side and an approximate weight on the other side, the weight of the object is determined. If the weight placed is more than the unknown weight, the weight is removed and another weight of smaller value is placed and again the measurement is performed. Now if it is found that the weight placed is less than that of the object, another weight of smaller value is added to the weight already present, and the measurement is performed. If it is found to be greater than the unknown weight the added weight is removed and another weight of smaller value is added. In this manner by adding and removing the appropriate weight, the weight of the unknown object is determined. The successive approximation DVM works on the same principle. Its basic block diagram is shown in Figure 3.10.

When the start pulse signal activates the control circuit, the successive approximation register (SAR) is cleared. The output of the SAR is 00000000. V_{out} of the D/A converter is 0. Now, if $V_{in} > V_{out}$ the comparator output is positive. During the first clock pulse, the control circuit sets the D_7 to 1, and V_{out} jumps to the half reference voltage.
The SAR output is 10000000. If V_{out} is greater than V_{in}, the comparator output is negative and the control circuit resets D_7. However, if V_{in} is greater than V_{out}, the comparator output is positive and the control circuits keeps D_7 set. Similarly, the rest of the bits beginning from D_7 to D_0 are set and tested. Therefore, the measurement is completed in 8 clock pulses.

Let $V_{in} = 1V$

Operation	D7	D6	D5	D4	D3	D2	D1	D0	V_{out} (V)	Compare	Output
D7 Set	1	0	0	0	0	0	0	0	2.5	$V_{in} < V_{out}$	D7 Reset
D6 Set	0	1	0	0	0	0	0	0	1.25	$V_{in} < V_{out}$	D6 Reset
D5 Set	0	0	1	0	0	0	0	0	0.625	$V_{in} > V_{out}$	D5 Set
D4 Set	0	0	1	1	0	0	0	0	0.9375	$V_{in} > V_{out}$	D4 Set
D3 Set	0	0	1	1	1	0	0	0	1.09375	$V_{in} < V_{out}$	D3 Reset
D2 Set	0	0	1	1	0	1	0	0	1.015625	$V_{in} < V_{out}$	D2 Reset
D1 Set	0	0	1	1	0	0	1	0	0.9765625	$V_{in} > V_{out}$	D1 Set
D0 Set	0	0	1	1	0	0	1	1	0.99609375	$V_{in} > V_{out}$	D0 Set

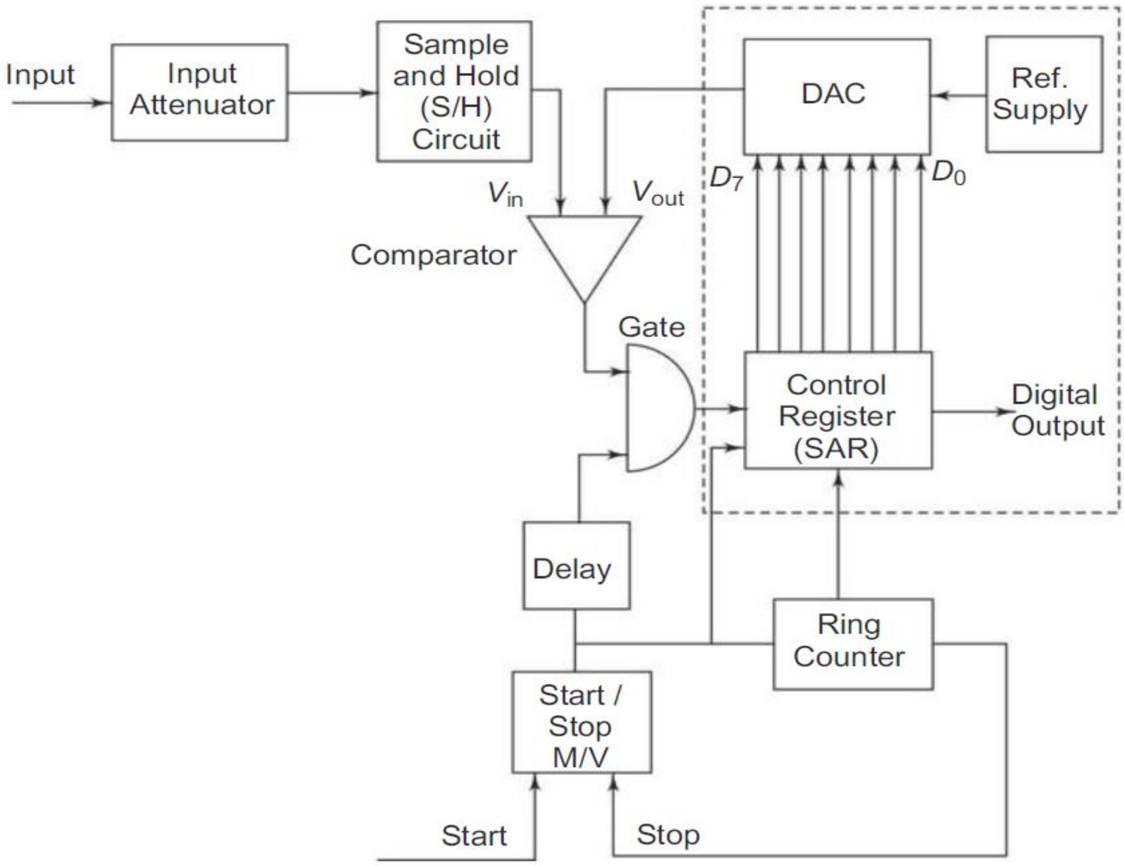

Figure 3.10 Successive Approximation DVM

At the beginning of the measurement cycle, a start pulse is applied to the start-stop multivibrator. This sets a 1 in the MSB of the control register and a 0 in all bits (assuming an 8-bit control) its reading would be 10000000. This initial setting of the register causes the output of the D/A converter to be half the reference voltage, i.e. 1/2 V. This converter output is compared to the unknown input by the comparator. If the input voltage is greater than the converter reference voltage, the comparator output produces an output that causes the control register to retain the 1 setting in its MSB and the converter continues to supply its reference output voltage of 1/2 V_{ref}.

The ring counter then advances one count, shifting a 1 in the second MSB of the control register and its reading becomes 11000000. This causes the D/A converter to increase its reference output by 1 increment to 1/4 V, i.e. 1/2 V + 1/4 V, and again it is compared with the unknown input. If in this case the total reference voltage exceeds the unknown voltage, the comparator produces an output that causes the control register to reset its second MSB to 0. The converter output then returns to its previous value of 1/2 V and awaits another input from the SAR. When the ring counter advances by 1, the third MSB is set to 1 and the converter output rises by the next increment of 1/2 V + 1/8 V. The measurement cycle thus proceeds through a series of successive approximations. Finally, when the ring counter reaches its final count, the measurement cycle stops and the digital output of the control register represents the final approximation of the unknown input voltage.

The S/H circuit is basically an Opamp that charges the capacitor during the Sample mode and retains the charge during the Hold mode.

DIGITAL INSTRUMENTS

3.10 Digital Tachometer

The technique employed in measuring the speed of a rotating shaft is similar to the technique used in a conventional frequency counter, except that the selection of the gate period is in accordance with the rpm calibration. Let us assume, that the rpm of a rotating shaft is R. Let P be the number of pulses produced by the pick up for one revolution of the shaft. Therefore, in one minute the number of pulses from the pick up is R x P. Then, the frequency of the signal from the pick up is (R x P)/60. Now, if the gate period is G s the pulses counted are (R x P x G)/60. In order to get the direct reading in rpm, the number of pulses to be counted by the counter is R. So we select the gate period as 60/P, and the counter counts:

$$\frac{(R \times P \times 60)}{60 \times P} = R \text{ pulses}$$

We can read the rpm of the rotating shaft directly. So, the relation between the gate period and the number of pulses produced by the pickup is G = 60/P. If we fix the gate period as one second (G = 1 s), then the revolution pickup must be capable of producing 60 pulses per revolution. Figure 3.15 shows a schematic diagram of a digital tachometer.

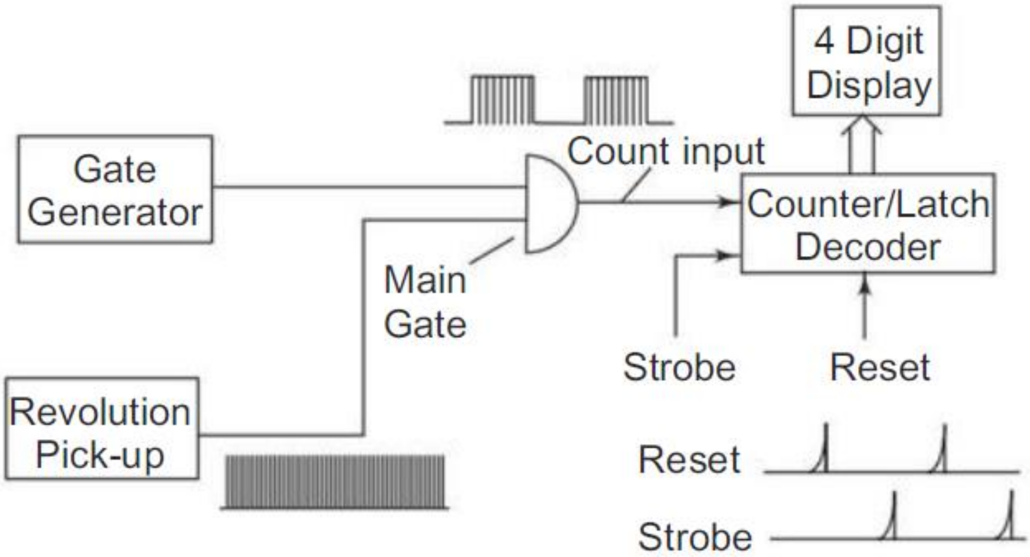

Figure 3.15 Basic block diagram of a Digital Tachometer

3.11 Digital pH Meter

The measurement of hydrogen ion activity (pH) in a solution can be accomplished with the help of a pH meter. For those unfamiliar with the terminology, a very brief review is included. pH is a quantitative measure of acidity. If the pH is less than 7, the solution is acidic (the lower the pH, the greater the acidity). A neutral solution has a pH of 7 and alkaline (basic) solutions have a pH greater than 7.

The pH unit is defined as, pH = – log (concentration of H+) where H+ is the hydrogen or hydronium ion. A digital pH meter differs from an ordinary pH meter, in this the meter is replaced by an analog to digital converter (ADC) and a digital display. A frequently used ADC for this application is the dual slope converter. A basic block diagram of a digital pH meter is shown in Figure 3.16.

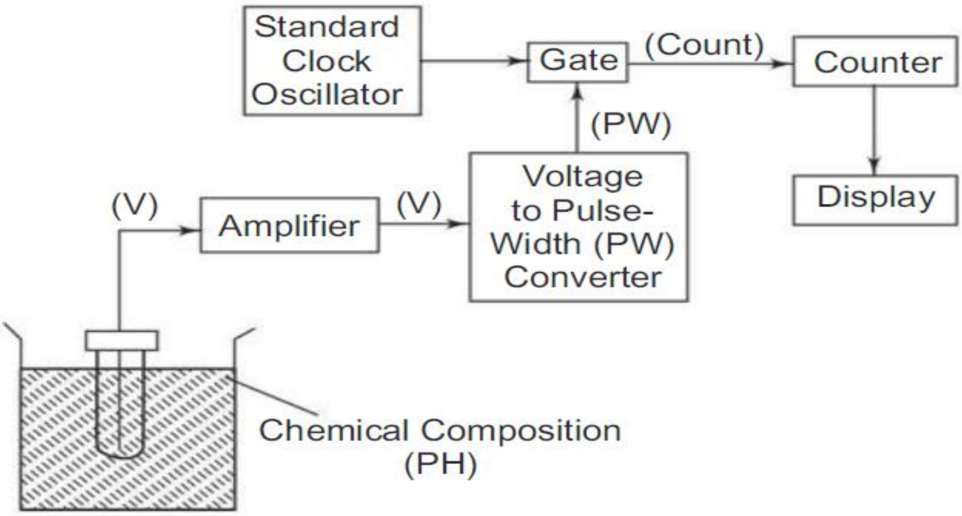

Figure 3.16 Digital pH Meter

The dual slope circuit produces a pulse which has a duration proportional to the input signal voltage, that is, a T pulse width signal. The pulse width is converted to a digital signal using the pulse to turn an oscillator On or Off, generating a count digital signal. The count signal is in turn counted or converted to a parallel digital signal for display by the counter.

3.12 Digital Phase Meter

The simplest technique to measure the phase difference between two signals employs two flip-flops. The signals to be fed must be of the same frequency. First, the signals must be shaped to a square waveform without any change in their phase positions, by the use of a zero crossing detector. The process of measuring the phase difference can be illustrated by the schematic diagram shown in Figure 3.17.

The block diagram consists of two pairs of preamplifier's, zero crossing detectors, J-K F/Fs, and a single control gate. Two signals having phases P_o and P_x respectively are applied as inputs to the preamplifier and attenuation circuit. The frequency of the two inputs is the same but their phases are different.

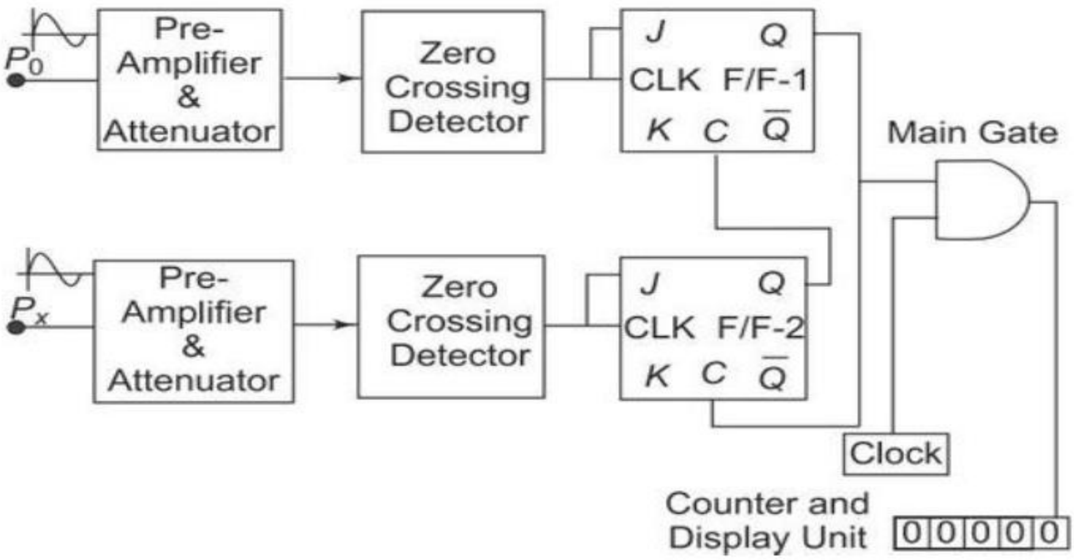

Figure 3.17 Digital phase meter

As the P_o input signal increases in the positive half cycle, the zero-crossing detector changes its state when the input crosses zero (0) giving a high (1) level at the output. This causes the J–K F/F–1 to be set (1), that is, the output (Q) of F/F–1 goes high. This high output from the F/F–1 enables the AND gate, and pulses from the clock are fed directly to the counter. The counter starts counting these pulses. Also, this high output level of F/F–1 is applied to the clear input of F/F–2 which makes the output of the F/F–2 go to zero (0).

Now as the input Px which has a phase difference with respect to Po, crosses zero (0) in the positive half cycle, the zero detector is activated, causing its output to go high (1). This high input in turn toggles the J–K F/F–2, making its output go high. This output (Q) of F/F–2 is connected to the clear input of F/F–1 forcing the F/F–1 to reset. Hence the output of F/F–1 goes to zero (0). The AND gate is thus disabled, and the counter stops counting. The number of pulses counted while enabling and disabling the AND gate is in direct proportion to the phase difference, hence the display unit gives a direct readout of the phase difference between the two inputs having the same frequency f. If the input signal frequency is f, then the clock frequency must be 360 times the input frequency for accurate measurements.